

A quadratic-Wasserstein Gaussian splitting estimate for the Bakry–Émery stability problem

Candidate full solution packet for arXiv:1807.09845

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Status. Candidate full solution, likely valid, requiring expert review. Under exactly the hypotheses of Courtade and Fathi’s Theorem 1.1, the argument below gives the missing W_2 estimate for every $k \leq n$, and in particular for $k < n$. The quantitative bound obtained is

$$W_2(\mu, \gamma_{p,E} \otimes \bar{\mu}) \leq 13k\sqrt{\varepsilon}.$$

1 Source question

The source is T. A. Courtade and M. Fathi, *Stability of the Bakry–Émery theorem on $\mathbb{R}^n$* [1]. It is arXiv:1807.09845, later published in *Journal of Functional Analysis* 279 (2020), 108523. Remark 1.1 on page 3 states:

When $k = n$, it is possible to improve the topology, and get an estimate of order $\sqrt{\varepsilon}$ in the stronger W_2 distance. The authors do not know how to get a W_2 estimate when $k < n$.

marginal distribution of μ on V^- , which enjoys the same convexity property as μ .

In this statement, W_1 stands for the classical L^1 Kantorovitch-Wasserstein distance [34], and $H^1(\mu) := \{f; \int (|f|^2 + |\nabla f|^2) d\mu < \infty\}$ is a weighted Sobolev space with respect to μ .

We shall obtain the following improvement:

Theorem 1.2. *Under the same assumptions as in Theorem 1.1, we have*

$$W_1(\mu, \gamma_{p,V} \otimes \bar{\mu}) \leq Ck^{3/2}\sqrt{\epsilon}.$$

In fact, we may take $C = 18\sqrt{2} < 26$.

Beyond the improved dependence on ϵ , the fact that our bound depends on k and not on n is useful for high-dimensional situations.

The proof of [23] is based on a stability version of Caffarelli's contraction theorem [11], which is a regularity estimate on the nonlinear Monge-Ampère PDE. To obtain the improved bound, we shall rely instead on Stein's method [32, 33], which is a way of quantifying distances between probability measures using well-chosen integration by parts formulas. See [31] for an introduction to the topic. The main reason why this allows us to obtain better estimates is that this proof mostly remains at a linear level, instead of relying on nonlinear tools as in [23]. The other main tool in the proof is that the test functions in the assumptions of the theorem can be viewed as approximate minimizers in a variational problem, which then give rise to an approximate Euler-Lagrange equation (up to reminder terms of order $\sqrt{\epsilon}$), which takes the form of an integration by parts formula. See Section 1.3 for an overview of the strategy of proof.

Remark 1.1. *When $k = n$, it is possible to improve the topology, and get an estimate of order $\sqrt{\epsilon}$ in the stronger W_2 distance, using results from [30, 21]. We do not know how to get a W_2 estimate when $k < n$, due to regularity issues for the Poisson equation we shall make use of in the proof.*

We do not know if the bound is optimal. Testing on Gaussian measures with variance $1 - \epsilon$ shows that the optimal rate cannot be better than ϵ (instead of $\sqrt{\epsilon}$). See also Remark 1.4 in [23] for computations in dimension one in a related problem which suggest the sharp rate could be ϵ .

Our main result has the following immediate corollary, which can be viewed as a dimension free improvement of the Bakry-Émery theorem

The hypotheses referred to in that remark are these. The probability measure $\mu = e^{-V} dx$ on $\mathbb{R}^n$ satisfies $\nabla^2 V \geq I_n$. There are $u_1, \dots, u_k \in H^1(\mu)$ such that

$$\int u_i d\mu = 0, \quad \int u_i^2 d\mu = 1, \quad \int \nabla u_i \cdot \nabla u_j d\mu = 0 \quad (i \neq j),$$

and

$$\int |\nabla u_i|^2 d\mu \leq 1 + \epsilon.$$

The desired product has a standard Gaussian factor on a k -plane E , with barycenter $p \in E$, and the other factor is the marginal $\bar{\mu}$ of μ on $E^\perp$.

2 Proof intuition

Let T be the Brenier map sending the standard Gaussian to μ . Caffarelli's contraction theorem says that its Jacobian $A = \nabla T$ is a symmetric positive contraction. Thus $B = I - A$ measures precisely the failure of T to be the identity in a direction.

The near-extremizers u_i pull back to near-extremizers of the Gaussian Poincaré inequality. Their first Hermite-chaos directions form a well-conditioned k -plane, and on that plane the Hilbert-Schmidt energy of B is $O(k\epsilon)$. This is the rigidity input.

The coupling is the second idea. Split a standard Gaussian as $G = Y + Z$ along the selected plane and independently resample only Y . Couple $T(Y + Z)$ to

$$p + Y + P_{E^\perp}T(Y' + Z),$$

where Y' is an independent copy of Y . The latter random vector has exactly the product law $\gamma_{p,E} \otimes \bar{\mu}$. Gaussian Poincaré controls both the direct displacement in E and the effect of resampling on the orthogonal component by the same Jacobian defect. This bypasses the Poisson-equation regularity obstruction mentioned in the source.

3 The result

Theorem 1 (Quadratic-Wasserstein splitting). *Let $\mu = e^{-V}dx$ be a probability measure on $\mathbb{R}^n$ with $\nabla^2 V \geq I_n$. Suppose that $u_1, \dots, u_k$ satisfy all of the normalization, gradient-orthogonality, and energy assumptions displayed above, for some $1 \leq k \leq n$ and $\varepsilon \geq 0$. Then there are a k -dimensional subspace $E \subset \mathbb{R}^n$ and $p \in E$ such that*

$$W_2(\mu, \gamma_{p,E} \otimes \bar{\mu}) \leq 13k\sqrt{\varepsilon},$$

where $\bar{\mu} = (P_{E^\perp})_\# \mu$.

We first isolate the coupling estimate.

Lemma 1 (Partial resampling coupling). *Let $T = \nabla \phi$ send the standard Gaussian γ_n to μ , and assume $0 \leq A := \nabla T \leq I$ almost everywhere. Let E be any k -plane, let $P = P_E$, put $B = I - A$, and set*

$$\delta_E := \mathbb{E} \|BP\|_{\text{HS}}^2.$$

If $p = \mathbb{E}[PT(G)]$, then

$$W_2^2(\mu, \gamma_{p,E} \otimes (P_{E^\perp})_\# \mu) \leq 3\delta_E.$$

In particular, $\delta_E \leq k$ and the left-hand side is at most $3k$.

Proof. Write $G = Y + Z$, where $Y \sim \gamma_E$ and $Z \sim \gamma_{E^\perp}$ are independent, and let Y' be an independent copy of Y , also independent of Z . Couple

$$X = T(Y + Z), \quad \tilde{X} = p + Y + P_{E^\perp}T(Y' + Z).$$

The two summands of $\tilde{X}$ are independent. Moreover, $Y' + Z \sim \gamma_n$, so

$$\text{Law}(\tilde{X}) = \gamma_{p,E} \otimes (P_{E^\perp})_\# \mu.$$

Apply the vector-valued Gaussian Poincaré inequality to $P(T(G) - G)$. Symmetry of A gives

$$\begin{aligned} \mathbb{E}|PT(G) - p - Y|^2 &\leq \mathbb{E}\|P(A - I)\|_{\text{HS}}^2 \\ &= \mathbb{E}\|(A - I)P\|_{\text{HS}}^2 = \delta_E. \end{aligned}$$

For fixed $Z = z$, apply the same inequality in the Y variable to $f_z(Y) = P_{E^\perp}T(Y + z)$. The identity for two independent copies and then Gaussian Poincaré give

$$\begin{aligned} \mathbb{E}|P_{E^\perp}T(Y + Z) - P_{E^\perp}T(Y' + Z)|^2 \\ = 2\mathbb{E}_Z \text{Var}_Y(f_Z(Y)) \leq 2\mathbb{E}\|P_{E^\perp}AP\|_{\text{HS}}^2 \leq 2\delta_E, \end{aligned}$$

because $P_{E^\perp}AP = -P_{E^\perp}BP$. The two displayed costs lie in orthogonal subspaces, so their sum is the squared cost of this coupling. Finally $0 \leq B \leq I$ implies $\delta_E \leq k$. $\square$

Proof of the theorem. Caffarelli's contraction theorem [2], in the form already used in the source paper, supplies the Brenier map T from γ_n to μ with $A = \nabla T$ symmetric and $0 \leq A \leq I$ almost everywhere. The standard approximation argument covers the stated Sobolev regularity; all estimates below are L^2 estimates.

First suppose

$$\varepsilon \leq \frac{1}{36k}. \quad (1)$$

Let $G \sim \gamma_n$, define

$$q_i = \nabla u_i(T(G)), \quad v_i = u_i \circ T,$$

and note that $\nabla v_i = Aq_i$. Since $\text{Var}_{\gamma_n}(v_i) = 1$, Gaussian Poincaré and the assumed energy bound imply

$$1 \leq \mathbb{E}|Aq_i|^2 \leq \mathbb{E}|q_i|^2 \leq 1 + \varepsilon.$$

For a positive contraction A , one has $(I - A)^2 \leq I - A^2$. Consequently,

$$\mathbb{E}|Bq_i|^2 \leq \mathbb{E}(|q_i|^2 - |Aq_i|^2) \leq \varepsilon. \quad (2)$$

Project v_i onto the first Gaussian chaos:

$$v_i(x) = w_i \cdot x + z_i(x),$$

where z_i is orthogonal to constants and linear functions. The improved Gaussian Poincaré inequality on that orthogonal complement yields

$$\mathbb{E}z_i^2 \leq \varepsilon, \quad \mathbb{E}|\nabla z_i|^2 \leq 2\varepsilon, \quad 1 - \varepsilon \leq |w_i|^2 \leq 1.$$

Indeed, $1 = |w_i|^2 + \mathbb{E}z_i^2$ and $\mathbb{E}|\nabla v_i|^2 = |w_i|^2 + \mathbb{E}|\nabla z_i|^2 \leq 1 + \varepsilon$, while $\mathbb{E}|\nabla z_i|^2 \geq 2\mathbb{E}z_i^2$.

Put $\widehat{w}_i = w_i/|w_i|$. Condition (1) gives $\sqrt{\varepsilon} \leq 1/6$. Combining (2), $Aq_i - w_i = \nabla z_i$, and $|w_i - \widehat{w}_i| = 1 - |w_i| \leq \varepsilon$, we obtain

$$\|q_i - \widehat{w}_i\|_{L^2(\gamma_n)} \leq \sqrt{\varepsilon} + \sqrt{2\varepsilon} + \varepsilon < \sqrt{7\varepsilon}, \quad (3)$$

$$\|B\widehat{w}_i\|_{L^2(\gamma_n)} \leq \|Bq_i\|_2 + \|B(\widehat{w}_i - q_i)\|_2 < \sqrt{14\varepsilon}. \quad (4)$$

The q_i are mutually orthogonal in $L^2(\gamma_n; \mathbb{R}^n)$, and each has norm at least one, because the Bakry–Émery Poincaré inequality gives $1 = \text{Var}_\mu(u_i) \leq \int |\nabla u_i|^2 d\mu$. Hence, for every $c \in \mathbb{R}^k$,

$$\begin{aligned} \left| \sum_i c_i \widehat{w}_i \right| &\geq \left\| \sum_i c_i q_i \right\|_2 - \left\| \sum_i c_i (q_i - \widehat{w}_i) \right\|_2 \\ &\geq (1 - \sqrt{7k\varepsilon})|c| \geq \frac{1}{2}|c|. \end{aligned}$$

Thus the $\widehat{w}_i$ span a k -plane E . If $W : \mathbb{R}^k \rightarrow E$ is defined by $Wc = \sum_i c_i \widehat{w}_i$, the last inequality says

$$WW^* \geq \frac{1}{4}P_E.$$

Using positivity and (4),

$$\delta_E = \mathbb{E} \text{tr}(B^2 P_E) \leq 4\mathbb{E} \text{tr}(B^2 WW^*) = 4 \sum_{i=1}^k \mathbb{E}|B\widehat{w}_i|^2 < 56k\varepsilon.$$

Lemma 1 now gives

$$W_2^2(\mu, \gamma_{p,E} \otimes \bar{\mu}) < 168k\varepsilon,$$

and therefore, since $k \geq 1$,

$$W_2(\mu, \gamma_{p,E} \otimes \bar{\mu}) < \sqrt{168k\varepsilon} \leq 13k\sqrt{\varepsilon}.$$

It remains to treat $\varepsilon > 1/(36k)$. Choose any k -plane E . The last assertion of Lemma 1 yields

$$W_2(\mu, \gamma_{p,E} \otimes \bar{\mu}) \leq \sqrt{3k} < 6\sqrt{3}k\sqrt{\varepsilon} < 13k\sqrt{\varepsilon}.$$

This completes all cases, including $k < n$. □

4 Verification and limitations

The proof uses only (i) Caffarelli contraction for the Gaussian-to- μ Brenier map, (ii) ordinary and second-chaos Gaussian Poincaré inequalities, and (iii) partial Gaussian resampling. The coupling has been checked directly: its second marginal is exactly the requested product, not an approximation. The two places most deserving expert scrutiny are the L^2 approximation passage for nonsmooth u_i and the operator trace inequality used to pass from the approximate directions to their span. Both are standard and are written explicitly in the proof.

The constant 13 is not claimed sharp. The source also asks whether the $\sqrt{\varepsilon}$ rate itself can be improved to order ε ; this packet does not answer that separate optimal-rate question.

The bounded novelty search (recorded in `novelty.md`) covered the run’s four lightweight indexes and primary arXiv searches using the exact title, the exact source sentence, “uniformly log-concave Poincaré stability W_2 ”, and “Gaussian factor splitting W_2 ”. It found the source paper and adjacent EPI, RCD, and interpolation results, but no later paper giving the stated $k < n$ estimate. This is not an exhaustive citation-graph review, so novelty confidence is moderate rather than definitive.

Human-review recommendation. Prioritize review. The argument is short, gives the exact missing topology under the exact source hypotheses, and appears to remove the stated regularity obstruction rather than assume it away.

References

- [1] T. A. Courtade and M. Fathi, *Stability of the Bakry–Émery theorem on $\mathbb{R}^n$* , arXiv:1807.09845; J. Funct. Anal. 279 (2020), 108523.
- [2] L. A. Caffarelli, *Monotonicity properties of optimal transportation and the FKG and related inequalities*, Comm. Math. Phys. 214 (2000), 547–563.